

Deep Generative Models

Lecture 4

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AI Masters

2026, Spring

Recap of Previous Lecture

MLE for LVM

$$\theta^* = \arg \max_{\theta} \sum_{i=1}^n \log p_{\theta}(\mathbf{x}_i) = \arg \max_{\theta} \sum_{i=1}^n \log \int p_{\theta}(\mathbf{x}_i | \mathbf{z}_i) p(\mathbf{z}_i) d\mathbf{z}_i.$$

Naive Monte Carlo Estimation

$$\log p_{\theta}(\mathbf{x}) = \log \mathbb{E}_{p(\mathbf{z})} p_{\theta}(\mathbf{x} | \mathbf{z}) \geq \mathbb{E}_{p(\mathbf{z})} \log p_{\theta}(\mathbf{x} | \mathbf{z}) \approx \frac{1}{K} \sum_{k=1}^K \log p_{\theta}(\mathbf{x} | \mathbf{z}_k),$$

where $\mathbf{z}_k \sim p(\mathbf{z})$.

Variational Lower Bound (ELBO)

$$\mathcal{L}_{q,\theta}(\mathbf{x}) = \mathbb{E}_q \log \frac{p_{\theta}(\mathbf{x}, \mathbf{z})}{q(\mathbf{z})}$$

$$\log p_{\theta}(\mathbf{x}) = \mathcal{L}_{q,\theta}(\mathbf{x}) + D_{\text{KL}}(q(\mathbf{z}) \| p_{\theta}(\mathbf{z} | \mathbf{x})) \geq \mathcal{L}_{q,\theta}(\mathbf{x}).$$

Recap of Previous Lecture

Variational Evidence Lower Bound (ELBO)

$$\log p_{\theta}(\mathbf{x}) = \mathcal{L}_{q,\theta}(\mathbf{x}) + D_{\text{KL}}(q(\mathbf{z})\|p_{\theta}(\mathbf{z}|\mathbf{x})) \geq \mathcal{L}_{q,\theta}(\mathbf{x}).$$

$$\mathcal{L}_{q,\theta}(\mathbf{x}) = \int q(\mathbf{z}) \log \frac{p_{\theta}(\mathbf{x}, \mathbf{z})}{q(\mathbf{z})} d\mathbf{z} = \mathbb{E}_q \log p_{\theta}(\mathbf{x}|\mathbf{z}) - D_{\text{KL}}(q(\mathbf{z})\|p(\mathbf{z}))$$

Log-likelihood Decomposition

$$\log p_{\theta}(\mathbf{x}) = \mathbb{E}_q \log p_{\theta}(\mathbf{x}|\mathbf{z}) - D_{\text{KL}}(q(\mathbf{z})\|p(\mathbf{z})) + D_{\text{KL}}(q(\mathbf{z})\|p_{\theta}(\mathbf{z}|\mathbf{x})).$$

- ▶ Rather than maximizing likelihood, maximize the ELBO:

$$\max_{\theta} p_{\theta}(\mathbf{x}) \quad \rightarrow \quad \max_{q,\theta} \mathcal{L}_{q,\theta}(\mathbf{x})$$

- ▶ Maximizing the ELBO with respect to the variational distribution q is equivalent to minimizing the KL divergence:

$$\arg \max_q \mathcal{L}_{q,\theta}(\mathbf{x}) \equiv \arg \min_q D_{\text{KL}}(q(\mathbf{z})\|p_{\theta}(\mathbf{z}|\mathbf{x})).$$

Recap of Previous Lecture

$$\begin{aligned}\mathcal{L}_{q,\theta}(\mathbf{x}) &= \mathbb{E}_q \log p_{\theta}(\mathbf{x}|\mathbf{z}) - D_{\text{KL}}(q(\mathbf{z})\|p(\mathbf{z})) = \\ &= \mathbb{E}_q \left[\log p_{\theta}(\mathbf{x}|\mathbf{z}) - \log \frac{q(\mathbf{z})}{p(\mathbf{z})} \right] d\mathbf{z} \rightarrow \max_{q,\theta}.\end{aligned}$$

Variational Posterior

$$\begin{aligned}q^*(\mathbf{z}) &= \arg \max_q \mathcal{L}_{q,\theta^*}(\mathbf{x}) = \\ &= \arg \min_q D_{\text{KL}}(q(\mathbf{z})\|p_{\theta^*}(\mathbf{z}|\mathbf{x})) = p_{\theta^*}(\mathbf{z}|\mathbf{x});\end{aligned}$$

Amortized Variational Inference

We restrict the family of possible distributions $q(\mathbf{z})$ to a parametric class $q_{\phi}(\mathbf{z}|\mathbf{x})$, **conditioned on data $\mathbf{x}$** and **parameterized by ϕ** .

Gradient Update

$$\begin{bmatrix} \phi_k \\ \theta_k \end{bmatrix} = \left[\begin{bmatrix} \phi_{k-1} + \eta \cdot \nabla_{\phi} \mathcal{L}_{\phi,\theta}(\mathbf{x}) \\ \theta_{k-1} + \eta \cdot \nabla_{\theta} \mathcal{L}_{\phi,\theta}(\mathbf{x}) \end{bmatrix} \right]_{(\phi_{k-1}, \theta_{k-1})}$$

Recap of Previous Lecture

$$\mathcal{L}_{\phi, \theta}(\mathbf{x}) = \mathbb{E}_{q_{\phi}(\mathbf{z}|\mathbf{x})} \log p_{\theta}(\mathbf{x}|\mathbf{z}) - D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z})) \rightarrow \max_{\phi, \theta}.$$

Gradient w.r.t. θ — Monte Carlo Estimation

$$\begin{aligned}\nabla_{\theta} \mathcal{L}_{\phi, \theta}(\mathbf{x}) &= \int q_{\phi}(\mathbf{z}|\mathbf{x}) \nabla_{\theta} \log p_{\theta}(\mathbf{x}|\mathbf{z}) d\mathbf{z} \approx \\ &\approx \nabla_{\theta} \log p_{\theta}(\mathbf{x}|\mathbf{z}^*), \quad \mathbf{z}^* \sim q_{\phi}(\mathbf{z}|\mathbf{x}).\end{aligned}$$

Gradient w.r.t. ϕ — Reparameterization Trick

$$\begin{aligned}\nabla_{\phi} \mathcal{L}_{\phi, \theta}(\mathbf{x}) &= \int p(\epsilon) \nabla_{\phi} \log p_{\theta}(\mathbf{x}|\mathbf{g}_{\phi}(\mathbf{x}, \epsilon)) d\epsilon - \nabla_{\phi} D_{\text{KL}} \\ &\approx \nabla_{\phi} \log p_{\theta}(\mathbf{x}|\mathbf{g}_{\phi}(\mathbf{x}, \epsilon^*)) - \nabla_{\phi} \text{KL}\end{aligned}$$

Variational Assumption

$$p(\epsilon) = \mathcal{N}(0, \mathbf{I}); \quad q_{\phi}(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\mu_{\phi}(\mathbf{x}), \sigma_{\phi}^2(\mathbf{x})).$$

$$\mathbf{z} = \mathbf{g}_{\phi}(\mathbf{x}, \epsilon) = \sigma_{\phi}(\mathbf{x}) \odot \epsilon + \mu_{\phi}(\mathbf{x}).$$

Recap of Previous Lecture

Training

- ▶ Pick a batch of samples $\{\mathbf{x}_i\}_{i=1}^B$ (here we use Monte Carlo technique).
- ▶ Compute the objective for each sample (apply the reparametrization trick):

$$\epsilon^* \sim p(\epsilon); \quad \mathbf{z}^* = \mathbf{g}_\phi(\mathbf{x}, \epsilon^*);$$

$$\mathcal{L}_{\phi, \theta}(\mathbf{x}) \approx \log p_\theta(\mathbf{x}|\mathbf{z}^*) - D_{\text{KL}}(q_\phi(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z})).$$

- ▶ Update parameters via stochastic gradient steps with respect to ϕ and θ .

Inference

- ▶ Sample $\mathbf{z}^*$ from the prior $p(\mathbf{z})$ ($\mathcal{N}(0, \mathbf{I})$);
- ▶ Generate data from the decoder $p_\theta(\mathbf{x}|\mathbf{z}^*)$.

Note: The encoder $q_\phi(\mathbf{z}|\mathbf{x})$ isn't needed during generation.

Recap of Previous Lecture

$$\mathcal{L}_{q,\theta}(\mathbf{x}) = \mathbb{E}_q \log p_{\theta}(\mathbf{x}|\mathbf{z}) - D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x})||p(\mathbf{z}))$$

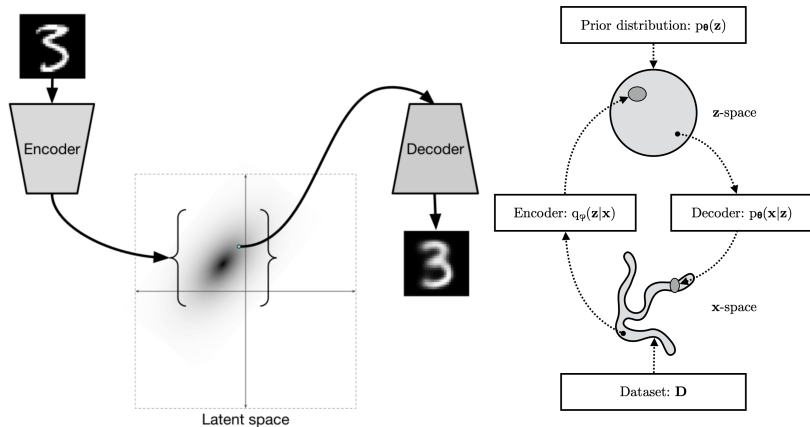


image credit: <http://ijdykeman.github.io/ml/2016/12/21/cvae.html>
Kingma D. P., Welling M. *An Introduction to Variational Autoencoders*, 2019

Outline

1. Discrete VAE Latent Representations
2. Vector Quantized VAE
3. ELBO Surgery

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Discrete VAE Latents

Motivation

- ▶ VAE has used **continuous** latent variables $\mathbf{z}$.
- ▶ For some modalities, **discrete** representations $\mathbf{z}$ may be a more natural choice.
- ▶ Advanced autoregressive models are highly effective for distributions over discrete variables (for example, transformers process discrete tokens).

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- ▶ Apply the reparametrization trick to obtain unbiased gradients.
- ▶ Use Gaussian distributions for $q_{\phi}(\mathbf{z}|\mathbf{x})$ and $p(\mathbf{z})$ to compute the KL analytically.

Discrete VAE Latents

Assumptions

- ▶ Let $c \sim \text{Cat}(\boldsymbol{\pi})$, where

$$\boldsymbol{\pi} = (\pi_1, \dots, \pi_K), \quad \pi_k = P(c = k), \quad \sum_{k=1}^K \pi_k = 1.$$

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$$= \sum_{k=1}^K q_{\phi}(k|\mathbf{x}) \log q_{\phi}(k|\mathbf{x}) - \sum_{k=1}^K q_{\phi}(k|\mathbf{x}) \log p(k) =$$

$$= -H(q_{\phi}(c|\mathbf{x})) + \log K.$$

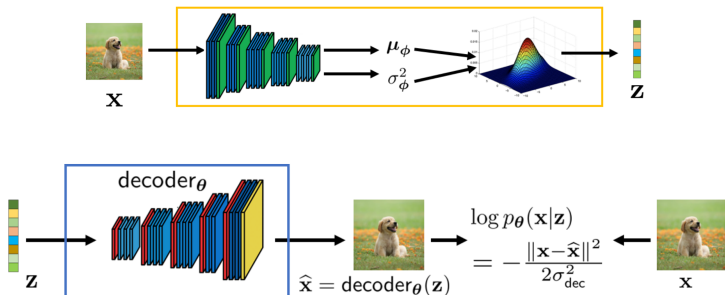
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- ▶ The encoder should output a discrete distribution $q_{\phi}(c|\mathbf{x})$.
- ▶ We need an analogue of the reparametrization trick for discrete $q_{\phi}(c|\mathbf{x})$.
- ▶ The decoder $p_{\theta}(\mathbf{x}|c)$ must take a discrete random variable c as input.



Outline

1. Discrete VAE Latent Representations
2. Vector Quantized VAE
3. ELBO Surgery

Vector Quantization

Define the codebook (dictionary) space $\{\mathbf{e}_k\}_{k=1}^K$ with $\mathbf{e}_k \in \mathbb{R}^L$ and K the number of codebook entries.

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Quantized Representation

A quantized vector $\mathbf{z}_q \in \mathbb{R}^L$, for any $\mathbf{z} \in \mathbb{R}^L$, is defined via nearest-neighbor lookup in the codebook:

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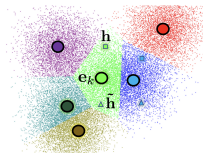
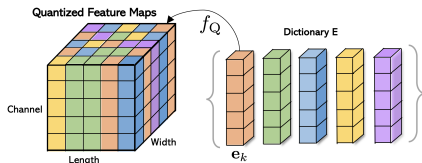
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Quantization Procedure

If the encoded tensor has spatial dimensions, quantization is independently applied to each of the $W \times H$ locations.



Vector Quantized VAE (VQ-VAE)

- ▶ The encoder outputs a continuous vector $\mathbf{z}_e = \text{NN}_{e,\phi}(\mathbf{x}) \in \mathbb{R}^L$.
- ▶ Quantization deterministically maps $\mathbf{z}_e$ to its quantized codebook vector $\mathbf{z}_q$.
- ▶ The decoder is conditioned on codebook entries $\mathbf{e}_c$, i.e., via $p_\theta(\mathbf{x}|\mathbf{e}_c)$ (instead of $p_\theta(\mathbf{x}|c)$).

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Deterministic Variational Posterior

$$q_\phi(c = k^*|\mathbf{x}) = \begin{cases} 1, & \text{for } k^* = \arg \min_k \|\mathbf{z}_e - \mathbf{e}_k\|; \\ 0, & \text{otherwise.} \end{cases}$$

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Note: The KL regularizer becomes constant and has no direct effect on the ELBO objective in this case.

Vector Quantized VAE (VQ-VAE): Forward

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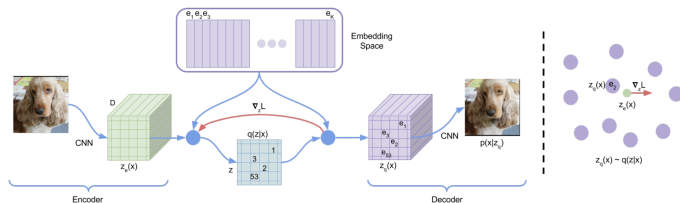
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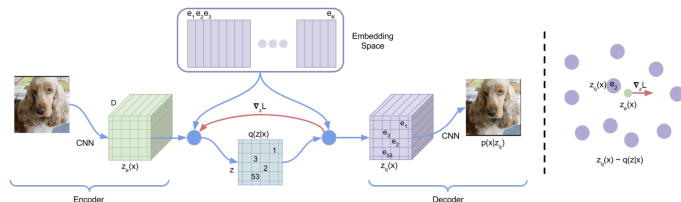
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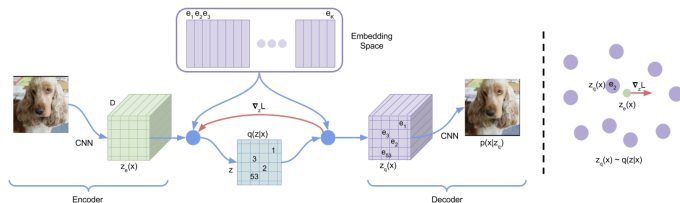
Challenge: The arg min operation is non-differentiable.

Vector Quantized VAE (VQ-VAE): Backward ELBO

$$\mathcal{L}_{\phi, \theta}(\mathbf{x}) = \log p_{\theta}(\mathbf{x}|\mathbf{z}_q) - \log K, \quad \mathbf{z}_q = \mathbf{e}_{k^*}, \quad k^* = \arg \min_k \|\mathbf{z}_e - \mathbf{e}_k\|.$$

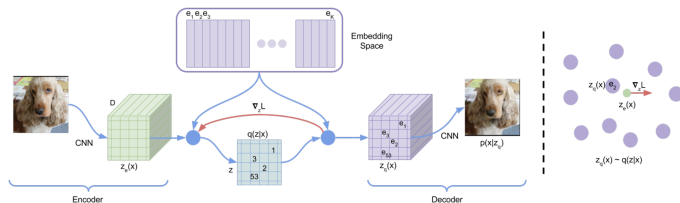
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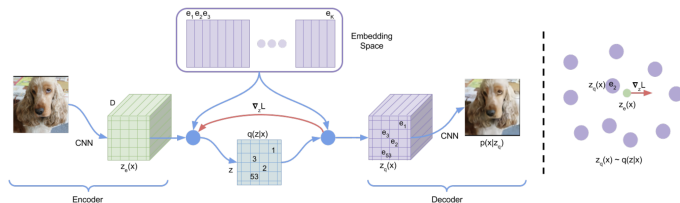


Straight-Through Gradient Estimator

$$\frac{\partial \log p(\mathbf{x} | \mathbf{z}_q, \theta)}{\partial \phi} = \frac{\partial \log p_{\theta}(\mathbf{x} | \mathbf{z}_q)}{\partial \mathbf{z}_q} \cdot \frac{\partial \mathbf{z}_q}{\partial \phi} =$$

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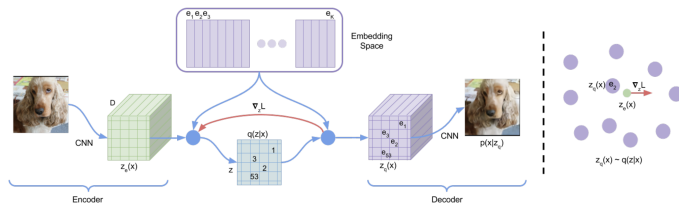


Straight-Through Gradient Estimator

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Straight-Through Gradient Estimator

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Vector Quantized VAE-2 (VQ-VAE-2)

Extension to the spatial domain: $\mathbf{c} \in \{1, \dots, K\}^{W \times H}$

$$q_{\phi}(\mathbf{c}|\mathbf{x}) = \prod_{i=1}^W \prod_{j=1}^H q(c_{ij}|\mathbf{x}, \phi); \quad p(\mathbf{c}) = \prod_{i=1}^W \prod_{j=1}^H \text{Uniform}\{1, \dots, K\}.$$

Sample Diversity



VQ-VAE (Proposed)

BigGAN deep

Vector Quantized VAE (VQ-VAE): Final algorithm

Training

- ▶ Vector-quantize (per spatial location if applicable):

$$k^* = \arg \min_k \|\mathbf{z}_e - \mathbf{e}_k\|_2, \quad \mathbf{z}_q = \mathbf{e}_{k^*}, \quad \mathbf{z}_e = \text{NN}_{e,\phi}(\mathbf{x}).$$

- ▶ Compute ELBO objective:

$$\mathcal{L}_{\phi,\theta}(\mathbf{x}) = \log p_{\theta}(\mathbf{x}|\mathbf{z}_q) - \log K.$$

- ▶ Compute total loss with codebook and commitment losses (with stop-gradient $\text{sg}[\cdot]$):

$$\mathcal{L} = -\mathcal{L}_{\phi,\theta}(\mathbf{x}) + \|\text{sg}[\mathbf{z}_e] - \mathbf{e}_{k^*}\|_2^2 + \beta \|\mathbf{z}_e - \text{sg}[\mathbf{e}_{k^*}]\|_2^2$$

- ▶ Use straight-through gradient estimation for encoder.

Vector Quantized VAE (VQ-VAE): Final algorithm

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- ▶ Vector-quantize (per spatial location if applicable):

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- ▶ Use straight-through gradient estimation for encoder.

Sampling

- ▶ Sample $c \sim p(c) = \text{Uniform}\{1, \dots, K\}$.
- ▶ Generate $\mathbf{x} \sim p_{\theta}(\mathbf{x}|\mathbf{e}_c)$.

Outline

1. Discrete VAE Latent Representations
2. Vector Quantized VAE
3. ELBO Surgery

ELBO Surgery

$$\frac{1}{n} \sum_{i=1}^n \mathcal{L}_{\phi, \theta}(\mathbf{x}_i) = \frac{1}{n} \sum_{i=1}^n \left[\mathbb{E}_{q_{\phi}(\mathbf{z}|\mathbf{x}_i)} \log p_{\theta}(\mathbf{x}_i|\mathbf{z}) - D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}_i) \| p(\mathbf{z})) \right].$$

ELBO Surgery

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Theorem

$$\frac{1}{n} \sum_{i=1}^n D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}_i) \| p(\mathbf{z})) = D_{\text{KL}}(q_{\text{agg}, \phi}(\mathbf{z}) \| p(\mathbf{z})) + \mathbb{I}_q[\mathbf{x}, \mathbf{z}];$$

ELBO Surgery

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- ▶ $q_{\text{agg}, \phi}(\mathbf{z}) = \frac{1}{n} \sum_{i=1}^n q_{\phi}(\mathbf{z}|\mathbf{x}_i)$ denotes the **aggregated** variational posterior.
- ▶ $\mathbb{I}_q[\mathbf{x}, \mathbf{z}]$ is the mutual information between $\mathbf{x}$ and $\mathbf{z}$ under the data distribution $p_{\text{data}}(\mathbf{x})$ and $q_{\phi}(\mathbf{z}|\mathbf{x})$.
- ▶ The first term encourages $q_{\text{agg}, \phi}(\mathbf{z})$ to match the prior $p(\mathbf{z})$.
- ▶ The second term reduces the information about $\mathbf{x}$ encoded in $\mathbf{z}$.

ELBO Surgery

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Proof

$$\frac{1}{n} \sum_{i=1}^n D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}_i) \| p(\mathbf{z})) = \frac{1}{n} \sum_{i=1}^n \int q_{\phi}(\mathbf{z}|\mathbf{x}_i) \log \frac{q_{\phi}(\mathbf{z}|\mathbf{x}_i)}{p(\mathbf{z})} d\mathbf{z}$$

ELBO Surgery

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ELBO Surgery

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ELBO Surgery

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ELBO Surgery

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ELBO Surgery

Revisiting the ELBO

$$\frac{1}{n} \sum_{i=1}^n \mathcal{L}_{\phi, \theta}(\mathbf{x}_i) = \frac{1}{n} \sum_{i=1}^n [\mathbb{E}_{q_{\phi}(\mathbf{z}|\mathbf{x}_i)} \log p_{\theta}(\mathbf{x}_i|\mathbf{z}) - D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}_i) \| p(\mathbf{z}))]$$

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The prior distribution $p(\mathbf{z})$ only appears in the last term.

ELBO Surgery

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Optimal VAE Prior

$$D_{\text{KL}}(q_{\text{agg}, \phi}(\mathbf{z}) \| p(\mathbf{z})) = 0 \quad \Leftrightarrow \quad p(\mathbf{z}) = q_{\text{agg}, \phi}(\mathbf{z}) = \frac{1}{n} \sum_{i=1}^n q_{\phi}(\mathbf{z}|\mathbf{x}_i).$$

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Hence, the optimal prior $p(\mathbf{z})$ is the aggregated variational posterior $q_{\text{agg}, \phi}(\mathbf{z})$.

Marginal KL

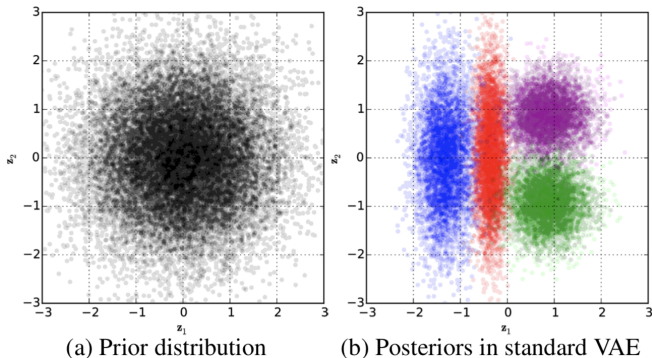
$$D_{\text{KL}}(q_{\text{agg},\phi}(\mathbf{z})\|p(\mathbf{z}))$$

- ▶ $q_{\phi}(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}_{\phi}(\mathbf{x}), \boldsymbol{\sigma}_{\phi}^2(\mathbf{x}))$ is unimodal.
- ▶ It is generally believed that the **mismatch between $p(\mathbf{z})$ and $q_{\text{agg},\phi}(\mathbf{z})$** is the primary explanation for blurry VAE-generated images.

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Why are VAE Generations Blurry?

Consider a Gaussian decoder $p_{\theta}(\mathbf{x}|\mathbf{z}) = \mathcal{N}(\mu_{\theta}(\mathbf{z}), \sigma^2 \mathbf{I})$.

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ELBO Optimization

With **fixed** encoder $q_{\phi}(\mathbf{z}|\mathbf{x})$, optimizing ELBO reduces to

$$\arg \min_{\theta} \mathbb{E}_{p_{\text{data}}(\mathbf{x})q_{\phi}(\mathbf{z}|\mathbf{x})} \|\mathbf{x} - \mu_{\theta}(\mathbf{z})\|^2$$

The optimal decoder mean is the **conditional expectation**:

$$\mu^*(\mathbf{z}) = \mathbb{E}_{q_{\phi}(\mathbf{x}|\mathbf{z})}[\mathbf{x}] = \frac{\mathbb{E}_{p_{\text{data}}(\mathbf{x})}[q_{\phi}(\mathbf{z}|\mathbf{x}) \cdot \mathbf{x}]}{\mathbb{E}_{p_{\text{data}}(\mathbf{x})}[q_{\phi}(\mathbf{z}|\mathbf{x})]}$$

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Blurriness from Averaging

- ▶ $p_{\text{data}}(\mathbf{x})$ and $q_{\phi}(\mathbf{z}|\mathbf{x})$ give us the aggregated posterior $q_{\text{agg},\phi}(\mathbf{z})$.
- ▶ If $\mathbf{x} \neq \mathbf{x}'$ map to overlapping latent regions, $\mu^*(\mathbf{z})$ averages over unrelated inputs.
- ▶ This leads to **blurry, non-distinct outputs**.

Summary

- ▶ Discrete VAE latents offer a natural class of latent variable models.
- ▶ Vector quantization provides a way to construct VAEs with discrete latents and deterministic variational posteriors.
- ▶ The straight-through gradient estimator allows gradients to pass as if quantization were an identity operation during backpropagation.
- ▶ ELBO surgery gives insights into the prior's influence in VAEs; the optimal prior is the aggregated variational posterior.
- ▶ The mismatch between $p(\mathbf{z})$ and $q_{\text{agg},\phi}(\mathbf{z})$ is widely regarded as the principal reason for VAE-generated image blurriness.